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MILITARY SPECIFICATION

TUBE UNIT ASSEMBLY, MEDICAL X-RAY APPARATUS

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers an X-Ray tube unit assembly for radiographic and fluoroscopic techniques, 100 PKV, 100 MA, 110/220 Volt, 50/60 cycles (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

V-T-295	Thread, Nylon.
GG-X-635	X-Ray Apparatus Set, Radiographic and Fluoroscopic, Medical.
TT-E-529	Enamel, Alkyd, Semi-Glass.
PPP-C-843	Cushioning, Material, Cellulosic.

Military

MIL-B-543	Buckles, Tongueless, and Web Strap.
MIL-C-496	Clips, End, Strap.
MIL-P-8585	Primer Coating, Zinc Chromate, Low-Moisture-Sensitivity.
MIL-W-530	Webbing and Tape, Textile, Cotton, General Purpose Natural or in Colors.
MIL-X-36068	X-Ray Apparatus Sets, Radiographic and Fluoroscopic, Medical (50/100 MA)

/FSC 6525/

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STANDARDS

Federal

FED-STD-72	Shockproof Cable Terminal and Receptacle For Use on X-Ray Equipment.
FED-STD-83	X-Ray Tube Focal Spot, Method of Measurement.
FED-STD-595	Colors.
FED-STD-751	Stitches, Seams and Stitchings.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWINGS

Defense Personnel Support Center (DPSC)

1774	Tube Unit Assembly, Medical X-Ray Apparatus, Rotating Anode, 100 MA.
10057	Modification of Packaging.
20653	Tube Insert, Medical X-Ray Apparatus, Rotating Anode.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

National Bureau of Standards (N.B.S.)
Handbook 76 - Medical X-Ray Protection Up to Three Million Volts.
(Applications for copies should be addressed to:
Superintendent of Documents, Washington 25, DC.

Underwriters Laboratories Inc. (UL)
UL 187 X-Ray Equipment
(Applications for copies should be addressed to:
207 East Ohio Street, Chicago Ill. 60611)

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3. REQUIREMENTS

3.1 First Article. Unless otherwise specified, prior to the start of production of the contract quantity, sample(s) of the Tube Unit Assembly shall be assembled, completely finished, and tested by the contractor. The sample(s) shall be submitted, together with the contractor's inspection and test reports, to the procuring contracting officer for examination and testing to determine compliance with the requirements of this specification. Written approval of the materials, construction, and performance of the sample shall be obtained from the procuring contracting officer, or his authorized representative, prior to the manufacture and assembly of the remaining quantity to be manufactured. Pending approval of the first article(s), acquisition of materials, components, or commencement of production is at the sole risk of the contractor. Accepted sample shall become the property of the procuring activity and shall be included in the quantity called for in the contract or order. In the event that the sample is not approved, the contractor shall submit a new sample for approval, or modify the existent sample so as to remove the deficiencies cited.

3.2 Fire and casualty hazards.

3.2.1 The contractor shall submit to the contracting agency prior to delivery proof that the equipment he will supply under this specification conforms to the requirements of Underwriters' Laboratories, Inc. (UL) Standard UL187 and the provisions of the National Electric Code. The label or listing of the UL may be accepted as evidence that the equipment conforms to this requirement.

3.2.2 In lieu of the label or listing, the contractor may submit prior to delivery independent proof satisfactory to the contracting agency that his equipment conforms to the published standards, including methods of tests, of the UL.

3.2.3 Compliance with UL requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure the acceptance of his commodity.

3.3 Radiation protection. The X-ray apparatus shall be constructed in accordance with requirements contained in the U.S. Department of Commerce, National Bureau of Standards Handbook 76, "Medical X-Ray Protection up to Three Million Volts." The lead lining with the tubehead housing shall confine leakage radiation to less than that allowed in Handbook No. 76 for diagnostic type protective tube housings.

3.3.1 Testing and Examination. All examination and testing of components shall be performed as indicated in paragraph 4.4.

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3.4 Material. Satisfactory high-grade materials shall be used throughout. All parts entering into the construction of the X-Ray tube unit assembly and chest inserts covered by this specification, shall be manufactured from materials which are entirely satisfactory for the purpose and of sufficient strength to maintain operational alignment for the life of the equipment.

3.4.1 Housing. The housing shall be fabricated of cast aluminum; and shall be suitably lined with lead to conform to the requirements of 3.3 for radiation.

3.4.2 Webbing. Webbing shall conform to MIL-W-530. Webbing shall be type IIA, 1 inch wide, vat dyed olive drab No. 7, and treated to be water-resistant and mildew-resistant.

3.4.3 Thread. Thread shall be nylon, conforming to V-T-295, for size E, type II, class 1, olive drab No. 7.

3.4.4 Buckles. Buckles shall conform to MIL-W-530 and shall be type II, class 1, 1 inch, a tongueless double bar, brass, with black oxidized finish.

3.4.5 Clips, end. End clips shall conform to MIL-C-496, and shall be type I, 1-1/2 inch, class 1.

3.4.6 Cushioning material. Cushioning material shall be made up of shredded foam latex bonded together with liquid latex; or rubberized curled hair sprayed with vinyl. The bottom and intermediate cushioning material shall have a density of eight pounds per cubic foot. The top cushioning material shall have a density of approximately 10 pounds per cubic foot.

3.4.7 Tray. The wood materials for the tray insert of the packing chest shall be as specified on Drawing 10057.

3.5 Stitches, seams and stitching. All required seams and stitching shall be straight and shall conform to stitch type 301 of Fed. Std. 751. The stitches shall be formed by machine and shall have a minimum of seven evenly spaced stitches per inch.

3.6 Components. Each tube unit assembly shall consist of the following major components:

A. X-ray tube unit (see 6.6).
B. Shockproof cables.
C. Rotor and fan cable.
D. Fitted packing chest.

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3.6.1 X-ray tube unit. The x-ray tube unit shall consist of an cone x-ray tubehead housing with an attached blower assembly, tube hanger, holder, filter slide and collapsible metal measuring ruler. The inner envelope of the housing shall be oil tight and shall contain the stator for the rotating anode, tube insert, and shockproof cable receptacles. The stator shall be held in place by a rigid support. The inner envelope shall be vacuum filled with insulating oil to prevent arcing and to conduct heat away from the tube insert. The tube hanger shall be provided for mounting the x-ray tube unit on a suitable tube stand (see 6.1).

3.6.1.1 Housing. The housing shall be shockproof and shall conform to Drawing 1774 for outside dimensions and configuration. The shockproof cable receptacles shall conform to the requirements in Federal Standard 72 for receptacle assemblies. The housing shall contain a collapsible metal measuring ruler which shall be so located as to indicate the distance from the end of its pointer to the focal spot of the x-ray tube. The ruler's scale shall include a range up to 72 inches and shall be graduated in inches and centimeters. The cone holder and filter slide shall be so mounted on the housing that the distance from the focal spot of the x-ray tube to the base of the cane, mounted in the cane holder, shall be 2-3/8 inches $\pm$ 1/32 inches. The tube hanger cone mount and filter slide shall be constructed in accordance with Drawing 1774. The blower assembly shall be enclosed within a sheet metal shroud which shall cover the x-ray tubehead housing. The turbine type blower wheel shall pull air into the blower housing and shall expel the air between the metal shroud and the x-ray tube housing. The temperature of the housing shall not exceed 203°F. when tested per 4.4.1. The cabling for the blower assembly shall conform to that of the schematic shown on Drawing 1774. The completely assembled x-ray tube housing, without cables, shall not weigh more than 40 pounds. A 1/2 millimeter (mm.) aluminum filter shall be provided for insertion into the filter mounting assembly.

3.6.1.2 X-ray tube insert. An x-ray tube shall be supplied which shall conform to Drawing 20653 for dimensions and configuration. The tube's envelope shall be clear glass to facilitate viewing the inner contents. The tube shall be a rotating anode double focus type with twin 1.5 mm focal spot sizes. The focal spot sizes shall be within the tolerances as specified in the test method shown in 4.4.4. The tube shall be designed to meet the following ratings (see 6.3):

a. Radiographic operation - 100 PKV, 100 MA for a minimum duration of 5.5 seconds.
b. Fluroscopic operation - 85 PKV and 5 MA.
c. Rotating anode acceleration to a minimum speed of 2400 vpm in 3/4 seconds.

The anode shall rotate in conjuction with a suitable stator winding that shall be supplied. The stator shall be designed to operate at 115 volts and shall have a starting current that shall not exceed 3.5 amperes.

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3.6.2 Shockproof cables. A pair of shockproof high-tension cables shall be supplied which shall conform to GG-X-635c, except that the overall length including high tension terminals shall be at least 16 feet.

3.6.2.1 Cable terminal waterpoofing kit. A cable waterproofing kit shall be supplied with the cables. The kit shall consist of a 2 ounce tube of silicone, or other similar waterproofing compound, a wiping cloth, and a wooden dowel. Instructions shall be included for the application of the waterproofing material to the cable terminals.

3.6.3 Rotor and fan cable. A low voltage 5 conductor cable shall be supplied to energize the blower assembly and the stator for the rotating anode. The cable shall be 16-1/2 feet long and shall conform to the cable schematic of Drawing 1774. The receptacle end of the cable shall be connected to the mating plug mounted in the blower assembly housing as shown in Drawing 1774.

3.6.4 Fitted packing chest. The x-ray tube head unit and shockproof cables shall be packed in a single chest. The chest shall be supplied to the contractor by the Government (see 6.4). The contractor shall supply inserts shaped to hold the tube unit and a wood tray to hold both the cables and bottom inserts. The fittings shall be fabricated in accordance with Drawing 10057. The surface of each cushion shall be sprayed with a thin layer of vinyl coating. When tested for compression in accordance with 4.4.5, the thickness of the cushion material after removal of load shall not be less than 92 percent of the original thickness. Corks or metal caps shall be permanently attached to the chest for the protection of the tubehead cable receptacles so as to exclude all dust and foreign matter from entering these receptacles during transportation. Rating charts conforming to the requirements of 3.12 shall be securely glued or cemented to the inside cover or lid of the chest. One side and one end of each chest shall be stencilled, in letters not less than 1/2-inch high, with the stock number and the item identification "TUBE UNIT ASSEMBLY, MEDICAL X-RAY APPARATUS, ROTATING ANODE, 100 MA 100 PKV." The side of the chest shall also be stencilled with the applicable contract number. One side shall be stencilled, "Re-usable Container." The four sides of the chest cover shall be stencilled, "For shipment and storage, this side up."

3.7 Interchangeability of parts. All individual parts and assemblies of parts of any complete unit supplied on a contract using this specification shall be interchangeable with the corresponding parts and assembly of parts or all other complete units supplied on that contract. In addition, major components of all tube units supplied on this specification shall be similarly interchangeable. This requirement shall also apply to chests inserts, which shall accommodate any tube unit meeting the requirements of this specification.

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Cooling rate (without blower assembly):

85° F.	12,500 H.U./min *
125° F.	8,000 H.U./min

* H.U. = Heat Units = KVP x ma. x time (in seconds)

The housing shall be air tight after being evacuated and then filled with oil. An acceptable unit, when subjected to the operational test of 4.4.1, shall show

1. stable operation
2. no indication of air leaks as evidenced by no air bubbles in the oil.

The x-ray tubehead shall have suitable expansion chambers to compensate for volumetric changes of oil which are caused by temperature changes within the range of -50° E to 125° F. The x-ray tube unit shall be capable of being stored and transported within the temperature range of -50° F. to 125° F, and shall successfully withstand the cold storage test as specified in 4.4.2. In addition, the x-ray tube unit shall be capable of performing satisfactorily for a minimum period of 500 consecutive hours when subjected to the breakdown test of 4.4.3.

3.8 Design and performance. The x-ray tube unit assembly shall be designed to operate at ambient temperatures between 32° F and 86° F at full ratings; and between 86° F and 125° F at reduced ratings. The thermal characteristics of the x-ray tube unit at 85° F and 125° F ambient temperatures shall be as follows:

Characteristic	Minimum ratings
Heat storage capacity:	
85° F ambient	1,000,000 H.U.*
125° F ambient	800,000 H.U.
Cooling rate (with blower assembly):	
85° F ambient	25,000 H.U./min.
125° F ambient	13,000 H.U./min.

3.9 Color. The color of the x-ray tube unit shall approximate color standard 24087 of Fed. Std. 595 (see 6.5).

3.10 Finish. The outside of the x-ray tube unit, except those surfaces which are stainless steel or plated, shall be finished with acid resisting, alkali resisting and solvent resisting enamel. Prior to the application of the finishing process, all surfaces to be enameled shall be thoroughly cleaned and chemically degreased. The finishing process, that issued, shall conform to the procedure outlined in 3.10.1.

3.10.1 Finishing process. The finishing process specified herein shall be applied in the sequence given:

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a. Primer. A zinc-chromate primer coat, conforming to MIL-P-8585, shall be applied to form a grounding bond and base for the finishing enamel.

b. Enamel. Two coats of sem-gloss enamel, conforming to TT-E-529 and the requirements of 3.9, shall be applied. Each coat, in accordance with the paint manufacturer's instructions, shall be sprayed on and baked in high temperature ovens. The resultant finish shall be hard smooth and highly resistant to abrasion, chipping, and cracking.

c. Clear enamel. A final coat of weather resisting, clear, synthetic baking enamel shall be applied and properly baked.

3.11 Identification marking. Each tube unit assembly shall be legibly and permanently marked with the following information:

Federal stock number
Manufacturer's name or registered trademark
Contract or purchase order number

In addition, the serial number and electrical characteristics shall appear. Marking secured by adhesives shall not be used.

3.11.1 Packing chest. The packing chest shall be marked in accordance with the requirements of 3.6.4.

3.12 Service data. The contractor shall furnish, with each unit, two copies of service data containing complete instructions for installation, operation maintenance, and a list of component parts. The service data shall include such detailed information which shall permit ready installation and maintenance of the item. All information contained in the manual shall be suitably indexed to permit ready reference to the required information. Wiring diagrams shall be included for electrical items or components. Wiring diagrams, drawings, and descriptive material shall include adequate information so that the replacement parts can be ordered by the manufacturer's part number, and without confusion as to the item identification. Identical or interchangeable replacement parts shall be identified by a single part number only. The service data shall indicate the focal spot sizes and ratings for the x-ray tube inserts that are supplied. Rating charts shall be included which shall indicate the KVP vs maximum exposure time for values of 25, 50 and 100 milliamperes (MA); for full wave rectified, half wave rectified, self rectified operation; for ambient temperatures of 85° F and 125° F; and for 50 and 60 cycle power lines. As soon as practicable after award of the contract or purchase order, the contractor shall furnish to the contracting officer, via the cognizant Government inspector, two preliminary copies of his proposed service data, for review and comment by Directorate of Medical Materiel, Defense Personnel Support Center. The preliminary service data shall be submitted sufficiently early to permit adequate review by the Directorate of Medical Materiel, Defense Personnel Support Center (based on a maximum period of 30 days for such review) and to allow incorporation by the contractor of required revisions and corrections in the final service data without delaying delivery under the contract or purchase order.

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One set of the service data, as indicated above, shall be placed in sealed, waterproof envelope, and affixed securely to the outside of the exterior container. The envelope shall be secured to the exterior container by use of metal ties, nails, or similar means, and shall be located at the end of the exterior container, preferably, when space permits, at the end bearing the markings required by MIL-STD-129. In cases where complete service data is too bulky to be conveniently attached to the outside of the exterior container, the contents of the exterior envelope may be limited to information necessary for roughing in preparations, and a note shall be made on the exterior envelope which shall indicate where the remainder of the first set of service data will be located. When two or more containers are required for packing a disassembled item (components of a single stock number), set markings shall be applied in accordance with MIL-STD-129. The envelope shall be affixed to the container designated as No. 1 of the set. The second set, and, if necessary, the remainder of the first set of service data, shall be attached to the unit inside the exterior container. In addition, the contractor shall mail 15 copies of the approved service data to the contracting officer for distribution. Should the supplier have furnished acceptable manuals for the identical item within two years preceding the data of contract, and propose to furnish manuals identical with those previously accepted, then the requirements for review of drafts and submission of 15 additional copies may be waived upon request.

3.13 Workmanship. Workmanship shall be first class throughout. The tube unit assemblies shall be free from defects which detract from their appearance of may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Supplier responsibilities for inspection. The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Total (major and minor combined) defects	II	4.0
For dimensional examination	S-3	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in table I.

Table I - Sampling Criteria

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Tube unit assembly	Operational test	3.6.1.1, 3.8	4.4.1	S-2
Tube unit assembly	Cold storage test	3.8	4.4.2	*
Tube unit assembly	Breakdown test	3.8	4.4.3	*
Tube insert	Focal spot size	3.6.1.2	4.4.4	*
Cushioning material	Compression set	3.6.4	4.4.5	S-2

* At option of the Government with one unit selected at random.

4.3 Examination. The x-ray tube units shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects:

Categories and defects *

Major

101 Components not properly aligned and fitted.
102 Metal parts not free of cracks, tears, ruptures, or deformed surfaces.
103 Evidence of oil leakage.
104 Service data missing or does not include all specified information.

Minor

201 Surfaces not free of burs, sharpness or sharp projections.
202 Finish not uniform.

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203 Unit not clean; not free of dirt, oil, or grease.
204 Identification marking missing, incomplete, illegible, or not permanently attached.
205 Ruler not operable or not as specified.

* Examination shall not be restricted to the classified possible defects listed above.

4.3.2 Dimensional examination. The x-ray tube units shall be examined for defects in dimensions. Any dimension not within the tolerances specified herein shall be classified as a defect.

4.4 Tests. All components to be tested shall be connected into, and made part of a complete x-ray apparatus set. Tests shall be performed in strict accordance with section 4 of Military Specification MIL-X-36068A and Section 4 of GG-X-635. Evidence or oil leakage within the tube unit housing shall be cause for rejection.

4.4.1 Operational test. A preliminary check shall be made of the housing for the presence of air or oil leakage. The x-ray tube head unit shall be installed as part of a suitable x-ray tube unit assembly, (see 6.1) which shall consist of a control unit and transformer as minimum components. Thermo-couple leads shall be attached to the tubehead housing. The x-ray tube unit assembly shall be operated in an ambient temperature of 85° F. Primary leads to high tension transformer shall be disconnected (i.e. no KVP). The x-ray tube unit shall be energized with filament voltage adjusted to light the filament. Observations shall be made through the x-ray window that:

1) No air bubbles present in the oil.
2) Tube rotor coming up to speed within one second.
3) The appropriate filament is energized by the desired positioning of the selector switch on the control unit.

In addition, the blower assembly shall be checked for proper operation. The x-ray tube unit assembly shall be deenergized and the primary lead shall be connected to the high tension transformer. The tube unit assembly, with blower assembly, shall be energized and shall be operated in the sequence outlined below. Sufficient time shall be allowed between exposures to dissipate heat from the tube. Reference shall be made to the manufacturer's rating charts and cooling curves for heat storage and cooling rates of the anode and tube.

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The sequence of operation shall be as follows:

a. Starting with 100 PKV and 2 MA; and gradual increase within 10 minutes to 110 PKV (current remaining constant).
b. 110 PKV and 2 MA for 8 minutes.
c. 85 PKV and 5 MA for 30 minutes.
d. 70 PKV and 50 MA for 5 seconds.
e. 100 PKV and 100 MA for 1 second repeated 5 times at 30 second intervals.
f. 100 PKV and 2 MA for 5 minutes.

The milliampere indicator or meter on the control unit shall be checked for stable operation of the tube.

4.4.2 Cold storage test. The x-ray tube unit shall be checked for the presence of air bubbles within the housing, by viewing through the x-ray window. The x-ray tube unit, outside its packing chest, shall be exposed to an ambient temperature between 65° F and 95° F for a minimum period of 24 hours. The x-ray tube unit shall again be tested for the presence of air bubbles. The x-ray tube unit shall then be placed in a cold chamber for a period of 18 hours with the temperature maintained at -50° F. The tube unit shall be removed from the cold chamber and shall be warmed by being placed in a room temperature of 65° F to 90° F for a period of 6 hours. A 10-inch fan shall be placed within 36 inches of the front of the tube unit, and shall provide the tube unit with air circulation during the warming period. After the 6 hour warming period the tube unit shall be examined for evidence of air within the housing. The cycle of exposure at -50° F for 18 hours, warming for 6 hours, and inspection, shall be repeated three times.

4.4.3 Breakdown test. The x-ray tube head unit shall be installed as part of a suitable x-ray tube unit assembly (see 6.1). The x-ray tube head unit shall be placed in an ambient temperature of 85° F. The control unit shall be set for a tube unit operation of 100 PKV and 3 MA. The x-ray tube unit assembly shall energized and shall be operated for a minimum period of 500 consecutive hours. The rotating anode and blower assembly shall also be in operation during the test.

4.4.4 Focal spot size. The tolerance on the focal spot size and its method of measurement shall be as specified in Fed. Std. No. 83.

4.4.5 Compression set. The cushioning material shall be subjected to the compression set test method specified in PPP-C-843a. The measured thickness and percentage compression set of the specimen shall be determined in accordance with PPP-C-843a.

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5. PREPARATION FOR DELIVERY

5.1 Packaging and packing.

5.1.1 Unit of issue. One x-ray tube unit, as specified, constitutes one unit of issue.

5.1.2 Level A. Each unit cushioned as specified in 3.4.6 and 3.4.7 shall be further packed in a goverment furnished container as specified in 3.6.4.

5.1.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.2 Marking.

5.2.1 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Special markings as applicable shall be applied as specified in 5.5.16.

6. NOTES

6.1 Intended use. This specification covers an X-ray tube unit assembly, component of Stock No. 6525-612-4715, X-Ray Apparatus Set, Radiographic and Fluoroscopic Medical, 100 MA, 1 Tube, 110-220 volt, 60 cycle, AC; and stock No. 6525-612-4720, X-Ray Apparatus Set, Radiographic and Fluoroscopic, Medical, 100 MA, 2 Tube 110-220 volt, 60 cycle, AC. The tube unit assembly shall be suitable for use with the following items:

Stock No.	Item Identification
6525-612-4725	CABLE ASSEMBLY and RADIOGRAPHIC CONE SET, MEDICAL X-RAY APPARATUS.
6525-612-4755	CONTROL UNIT, X-RAY APPARATUS, 100 MA, 100 PKV, 110-220- volt, 60 cycles, AC.
6525-612-4770	TABLE, RADIOGRAPHIC, Tilting, 110-220 volt, 60 cycle, AC.
6525-612-4775	TRANSFORMER, X-RAY APPARATUS, 100 MA, 100 PKV 110-220 volt, 60 cycle, AC.
6525-612-4790	TUBE STAND UNIT, MEDICAL X-RAY APPARATUS, Rail Mounted.
6525-615-8100	TABLE, FIELD X-RAY APPARATUS.

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Stock No.	Item Identification
6525-684-8745	X-RAY APPARATUS SET, RADIOGRAPHIC AND FLUOROSCOPIC, MEDICAL, Field, Mobile 50 MA 90 PKV, 110-220 volt, 50-60 cycle, AC.
6525-684-8780	X-RAY APPARATUS SET RADIOGRAPHIC AND FLUOROSCOPIC, MEDICAL, Field, 50 MA, 90 PKV, 110-220 volt, 50-60 cycle, AC.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing (see 5.1 and 5.1.3).
(c) Where to send preproduction sample (see 3.1).

6.3 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.4 The x-ray tube inserts that conform to this specification are Dunlee Part No. RZ2, Machlett Part No. Dynamax 30, Eureka Part No. RAD-T, or equal (see 3.6.1.2).

6.4.1 The Government furnished field chest shall conform to the requirements for FSN 6545-914-3510, chest, Medical Instrument and Supply Set, Field, No. 6, 30 inches long by 18 inches wide by 20 inches deep empty.

6.5 Availability of color chips. The individual 3-by5-inch color chips of the color referenced in 3.9 are available from General Services Administration, Business Service Center, Region 3, Seventh and D Streets, N.W. Washington 25 D.C. Price 5 cents.

6.6 The tube unit and tube insert are available as repair parts:

Federal Stock Number - 6525-612-8643 - TUBE UNIT, MEDICAL X-RAY APPARATUS, RADIOGRAPHIC AND FLUOROSCOPIC, ROTATING ANODE, 100 MA

Federal Stock Number - 6525-612-4780 - TUBE INSERT, MEDICAL X-RAY APPARATUS, ROTATING ANODE.

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Custodians:
Army - MD
Navy - MS
Air Force - 03

Review activities:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

CATAGORY N

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document (s) or serve to amend contractual requirements.

SPECIFICATION

MIL-T-36475A

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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